ORBIT

Voice-controlled browser automation for more natural, accessible web interaction

AI Agent / Voice-Controlled Browser Automation
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THE PROBLEM

Traditional browser use is a sequence of repetitive manual operations: finding controls, navigating pages, typing into fields, scrolling, and extracting information. This is time-consuming, difficult to repeat consistently, and excludes users who prefer voice-based interaction. Everyday browsing still lacks a natural-language interface that can translate intent into safe, verifiable browser actions.

THE PROPOSED SOLUTION

ORBIT turns a spoken request into a controlled browser workflow. It captures voice, converts speech to text, understands the user's intent, plans one safe action at a time, identifies visible page elements, executes the action, observes the result, and continues until the task is verified. For conversational requests, it can respond without touching the page.

SYSTEM ARCHITECTURE

The extension supplies the current page's compact DOM observation and optional screenshot to the backend planner. Validated actions are executed through Chrome extension APIs and content scripts; the loop is bounded and supports cancellation and confirmation for consequential actions.

KEY FEATURES

- Implemented: voice-controlled interaction, natural-language commands, AI command understanding, navigation, DOM-based element identification, clicking, typing, scrolling, extraction, task verification, and voice replies.
- Implemented: floating Chrome interface, screenshot-assisted interaction, persistent panel state, and controlled action vocabulary.
- Planned: richer workflow memory, broader integrations, and deeper task-specific capabilities beyond the current validated browser actions.

TECHNOLOGY STACK

- Chrome Manifest V3 extension with content scripts, service worker, storage, tabs, scripting, and debugger APIs.
- Node.js 18+ / Express backend with validated JSON action contracts.
- Gnani Prisma v3.0 via `/stt/v3` for speech-to-text.
- Gnani Timbre v2.5 via `/api/v1/tts/inference` for WAV voice output.
- OpenAI-compatible planner API configured for OpenRouter; the codebase also retains provider compatibility for LM Studio and Ollama.

INNOVATION & IMPACT

Natural interaction
Voice becomes the command layer for browser tasks, reducing reliance on repetitive pointing and typing.

Accessible by design
A conversational interface can make everyday web interaction more approachable for voice-first users.

Extensible agent loop
The observe → plan → act → verify pattern provides a foundation for richer, safe AI workflows.

DEMONSTRATION The accompanying demonstration video showcases ORBIT performing real browser automation through voice commands.